

InteliLite 9

Controller for single gen-set applications

SW version 1.0.0

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1 Front panel elements

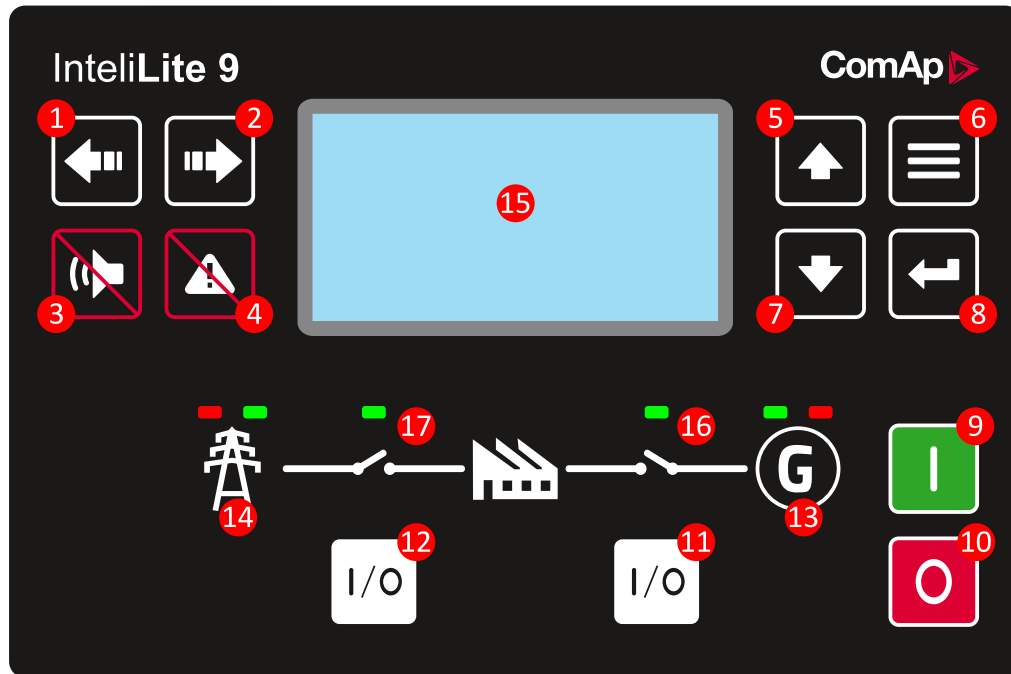


Image 1.1 Operator interface of IntelliLite 9

Control buttons		
Position	Picture	Description
1		LEFT button. Use this button to move left or to change the mode. The button can change the mode only if the main screen with the indicator of currently selected mode is displayed. <i>Note: This button will not change the mode if the controller mode is forced by one of binary inputs listed in the Reference Guide – “Operating modes” chapter.</i>
2		RIGHT button. Use this button to move right or to change the mode. The button can change the mode only if the main screen with the indicator of currently selected mode is displayed. <i>Note: This button will not change the mode if the controller mode is forced by one of binary inputs listed in the Reference Guide – “Operating modes” chapter.</i>
3		HORN RESET button. Use this button to deactivate the horn output without acknowledging the alarms.
4		FAULT RESET button. Use this button to acknowledge alarms and deactivate the horn output. Inactive alarms will disappear immediately and status of active alarms will be changed to "confirmed" so they will disappear as soon as their reasons dismiss.

5		UP button. Use this button to move up or increase value.
6		PAGE button. Use this button to switch over display pages.
7		DOWN button. Use this button to move down or decrease value.
8		ENTER button. Use this button to finish editing a setpoint or moving right in the history page.
9		START button. Works in MAN mode only. Press this button to initiate the start sequence of the engine.
10		STOP button. Works in MAN mode only. Press this button to initiate the stop sequence of the gen-set. Repeated pressing of button will cancel current phase of stop sequence (like cooling) and next phase will continue.
11		GCB button. Works in MAN and TEST modes only. Press this button to open or close the GCB.
12		MCB button. Works in MAN and TEST modes only. Press this button to open or close the MCB.

Indicators and others

Position	Description
13	GENERATOR status indicator. There are two states - Gen-set OK (indicator is green) and Gen-set failure (indicator is red). Green LED is on if the generator voltage is present and within limits. Red LED starts flashing when gen-set failure occurs. After FAULT RESET button is pressed, goes to steady light (if an alarm is still active) or is off (if no alarm is active).
14	MAINS status indicator. There are two states - Mains OK (indicator is green) and Mains failure (indicator is red). Green LED is on, if mains is present and within limits. Red LED starts blinking when the mains failure is detected and after the gen-set has started and connected to the load it lights permanently until the mains failure disappears.
15	Graphic B/W display, 132x64 pixels.
16	GCB ON . Green LED is on if GCB is closed. It is driven by GCB CLOSE/OPEN output or by GCB feedback signal.
17	MCB ON . Green LED is on if MCB is closed. It is driven by MCB CLOSE/OPEN output or by MCB feedback signal.

2 Display screens and pages structure

The displayed information is structured into "pages" and "screens". Use PAGE button to switch over the pages.

- ▶ The page Measurement consists of screens which display measured values like voltages, current, oil pressure etc., computed values like i.e. gen-set power, statistic data and the alarm list on the last screen.
- ▶ The page Setpoints contains all setpoints organized to groups and also a special group for entering password.
- ▶ The page History log shows the history log in the order that the last record is displayed first.

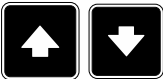
2.1 Main Screen



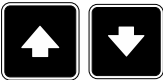
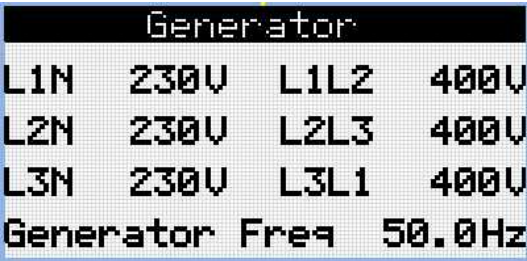
2.1.1 Symbols

- ▶ Padlock - active when LBI ACCESS LOCK is active
- ▶ R - active when there is active remote connection to controller
- ▶ Exclamation mark - active when there is any alarm in alarmlist

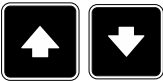
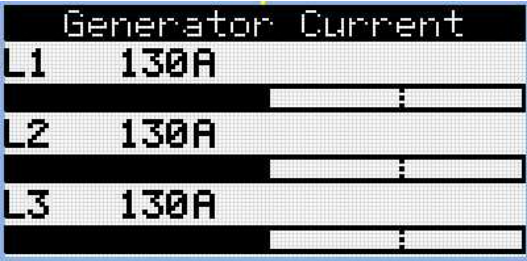
2.2 Measurement Screens



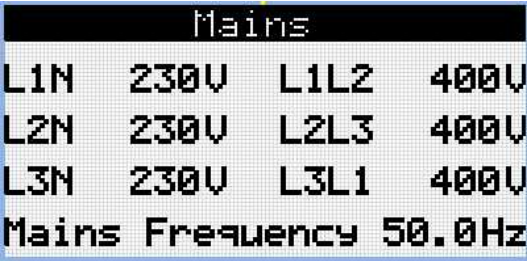
Note: Use Up and Down button to move between measurement pages.



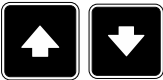
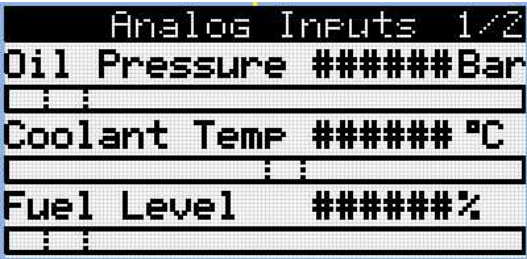
Note: Use Up and Down button to move between measurement pages.



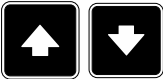
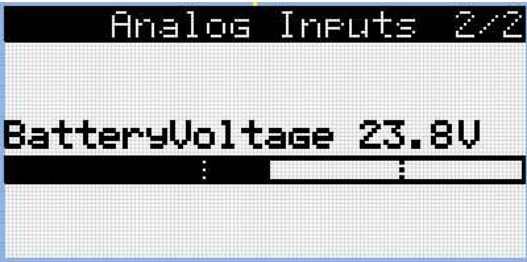
Note: Use Up and Down button to move between measurement pages.



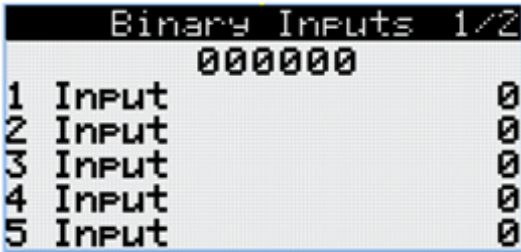
Note: Use Up and Down button to move between measurement pages.



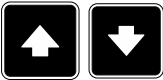
Note: Use Up and Down button to move between measurement pages.



Note: Use Up and Down button to move between measurement pages.

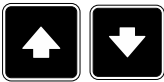


Note: Use Up and Down button to move between measurement pages.



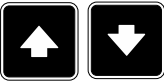
Note: Use Up and Down button to move between measurement pages.

Binary Outputs 1/2		
000000		
1	Output	0
2	Output	0
3	Output	0
4	Output	0
5	Output	0



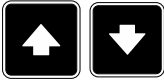
Note: Use Up and Down button to move between measurement pages.

Binary Outputs 2/2		
000000		
6	Output	0



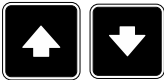
Note: Use Up and Down button to move between measurement pages.

EM-BIOS-EFCP 1/2		
10101010		
IN:	Input	1
	Not Used	0
OUT:	Output	1
IN:	Input	0
OUT:	Output	1



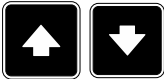
Note: Use Up and Down button to move between measurement pages.

EM-BIOS-EFCP 2/2		
00101010		
OUT: Output		0
OUT: Output		1
OUT: Output		0



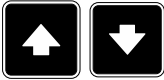
Note: Use Up and Down button to move between measurement pages.

Gen-Set Power			
	kW	PF	kVA
L1	0	0.000	0
L2	0	0.000	0
L3	0	0.000	0
Σ	0	0.000	0



Note: Use Up and Down button to move between measurement pages.

ECU Values	
Fuel Rate	#####l/h
CoolantTemp	#####°C
IntakeTemp	#####°C
Oil Press	#####bar
Boost Press	#####bar
Load	#####%



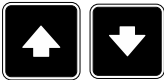
Note: Use Up and Down button to move between measurement pages.

Statistics 1/2	
Genset kWh	0
Genset kVArh	0
Mains kWh	0
Mains kVArh	0
Running Hours	0
Num Starts	0



Note: Use Up and Down button to move between measurement pages.

Statistics 2/2	
Num E-Stops	0
Shutdowns	0
Maintenance 1	0
Maintenance 2	0
Maintenance 3	0



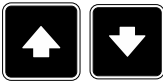
Note: Use Up and Down button to move between measurement pages.

CM-4G-GPS 1/2	
Cell Signal Lev	93%
Cell ErrorRate	12%
Cell Status	/
Cell Diag Code	12
Operator	T-Mobile CZ
Connection Type	4G



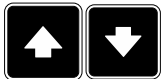
Note: Use Up and Down button to move between measurement pages.

CM-4G-GPS 2/2	
Latitude	0.1234
Longitude	0.1234
Altitude	123m
HomePosDist	123km
Satelites	1

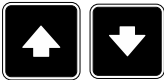
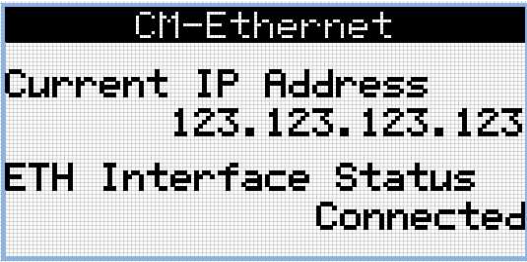


Note: Use Up and Down button to move between measurement pages.

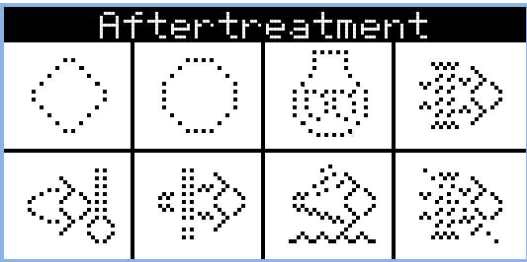
CM-GPRS	
Cell Signal Lev	93%
Cell ErrorRate	12%
Cell Status	/
Cell Diag Code	12
Operator	T-Mobile CZ



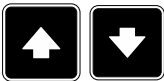
Note: Use Up and Down button to move between measurement pages.



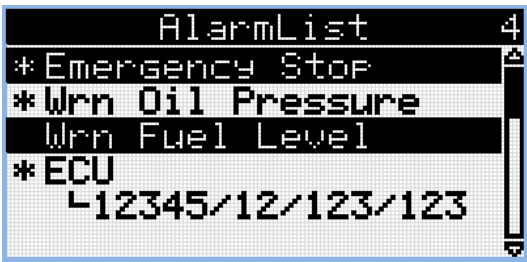
Note: Use Up and Down button to move between measurement pages.



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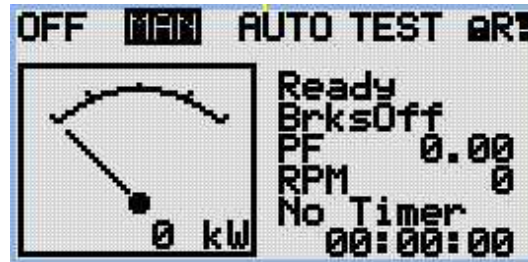


Note: Use Up and Down button to move between measurement pages.

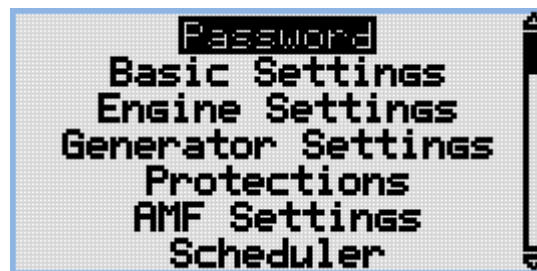
Note: From all of these pages it is possible to switch seamlessly to the setpoint group page by pressing Page button.

Note: There can be some additional screens and also some screens can be hidden. Screen's visibility depends on actual configuration (usage of extension or communication modules, ECU, etc.).

2.3 Setpoint Screens



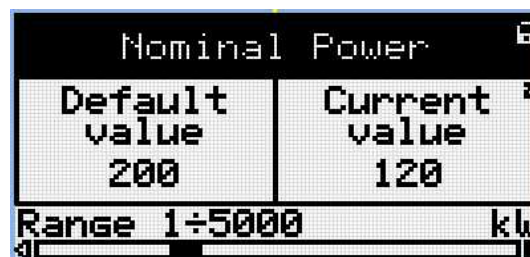
Note: From all measurement pages we can fluently go to the setpoint group page by pressing Page button.



Note: Use Up and Down button to select required setpoint group.



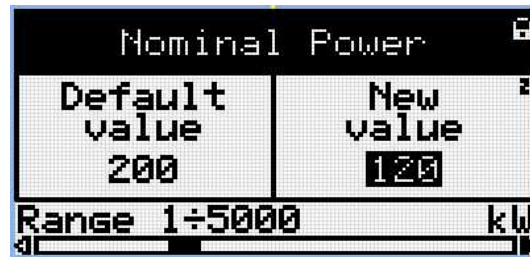
Note: Use Enter button to enter selected setpoint group.



Note: Use Left and Right button to select required setpoint.



Note: Use Enter button to enter selected setpoint.



Note: Use Up and Down button to set required value of selected setpoint.



Note: Use Enter button to confirm adjusted value of setpoint.



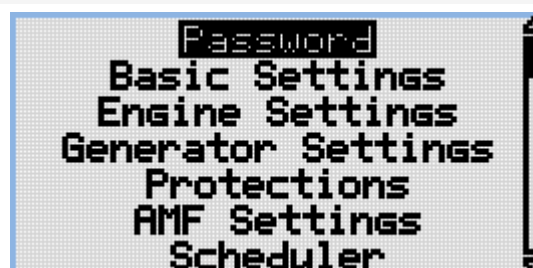
Note: Use Page button to discard changes, to set setpoint to previous value and to return to the list of setpoints of selected group.

IMPORTANT: Cannot change setpoint? Setpoints marked with an padlock are password protected. Enter password as described in the chapter Password (page 18).

2.4 History Log



Note: From all measurement pages we can fluently go to the setpoint group page by pressing Page button.



Note: From setpoint group page we can fluently go to the history log pages by pressing Page button.

No.	Reason
-001	GCB Closed
-002	MCB Opened
-003	MCB Closed
-004	GCB Opened
-005	GCB Closed
11:05:45	15/03/2014



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

Time	Date
14:01:43AM	15/03/2014
11:05:45AM	15/03/2014
11:01:43AM	15/03/2014
11:04:43AM	14/03/2014
14:41:43AM	11/03/2014
-002	MCB Opened



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

RPM	Pwr	Q
1500	15.0	15.0
0	0.0	0.0
0	0.0	0.0
1500	15.0	15.0
1500	15.0	15.0
-002	MCB Opened	



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

PF	LChr	GFra
0.75	C	50.0
0.00	C	0.0
0.00	C	0.0
0.73	C	50.0
0.74	C	50.0
-002	MCB Opened	



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

U _{G1}	U _{G2}	U _{G3}
230	230	230
0	0	0
0	0	0
230	230	230
230	230	230
-002 MCB Opened		



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

U _{G12}	U _{G23}	U _{G31}
230	230	230
0	0	0
0	0	0
230	230	230
230	230	230
-002 MCB Opened		



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

I _{L1}	I _{L2}	I _{L3}
30	30	30
0	0	0
0	0	0
30	30	30
30	30	30
-002 MCB Opened		



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

U _{m1}	U _{m2}	U _{m3}
230	230	230
0	0	0
0	0	0
230	230	230
230	230	230
-002 MCB Opened		



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

Um12	Um23	Um31
230	230	230
0	0	0
0	0	0
230	230	230
230	230	230
-002 MCB Opened		



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

FC	FMI
0	0
0	0
0	0
0	0
0	0
-002 MCB Opened	



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

MFrq	UBat	OilP
50.0	23.2	3.2
0.0	0.0	0.0
0.0	23.2	0.0
50.0	23.3	3.2
50.0	23.3	3.2
-002 MCB Opened		



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

EngT	FLvl
30.0	50
22.0	20
23.0	30
23.0	30
23.0	50
-002 MCB Opened	



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

EIN	EOU
011010	110000
110010	010010
010101	010101
110100	011010
110000	010101
-002 MCB Opened	



Note: Use Up and Down button to select required alarm reason.



Note: Use Enter button to move to the next page of history log.

Mode	
MAN	
MAN	
MAN	
MAN	
MAN	
-002 MCB Opened	



Note: Use Up and Down button to select required alarm reason.

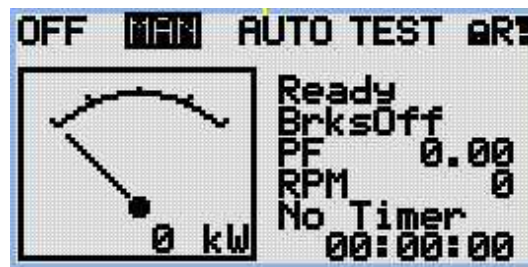


Note: Use Enter button to move to the first page of history log.

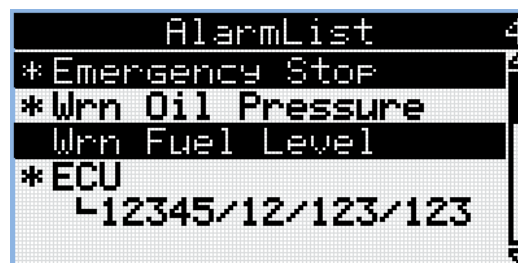
IMPORTANT: The records are numbered in reverse order, i.e. the latest (newest) record is "0" and older records have "-1", "-2" etc.

Note: This is only basic history record. There can be some additional screens in case that in controller is extension module or ECU is configured. Also it depends on connection type.

3 Browsing alarms



Note: Use Up button to move to alarmlist from main measurement screen.

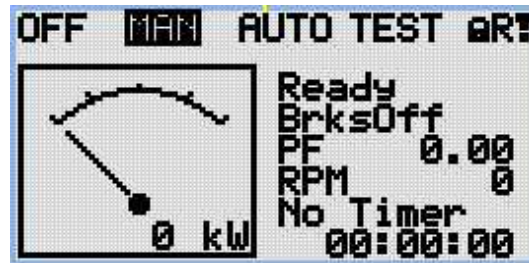


- ▶ Active alarms are displayed as white text on black background. It means the alarm is still active, i.e. the appropriate alarm conditions are still present.
- ▶ Inactive alarms are displayed as black text on white background. It means the alarm is no more active, i.e. the appropriate alarm conditions are gone.
- ▶ Not confirmed alarms are displayed with an asterisk. It means the alarm is still not acknowledged (confirmed).
- ▶ ECU alarms: SPN/FMI/OC/SC
 - SPN - Suspect parameter number
 - FMI - type of protection
 - OC - number of errors
 - SC - source of error

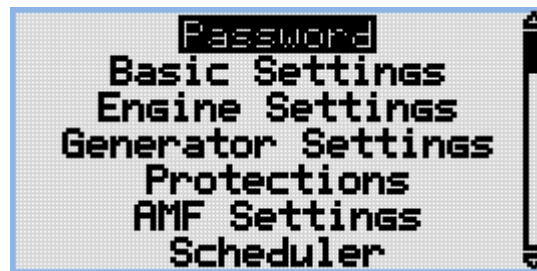


4 Password

4.1 Enter password



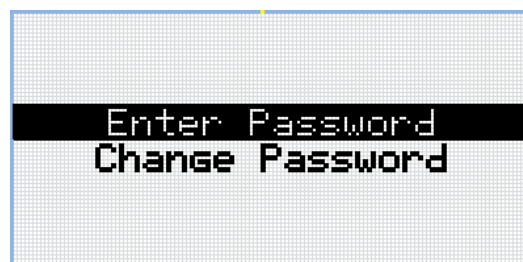
Note: From all measurement pages we can fluently go to the setpoint group page by pressing Page button.



Note: Use Up and Down button to select setpoint group Password.



Note: Use Enter button to enter setpoint group Password.



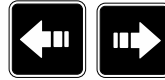
Note: Use Up and Down button to select Enter Password.



Note: Use Enter button to enter selected setpoint.



Note: Use Up and Down button to set required value of selected setpoint.



Note: Use Left and Right button to move between digits.

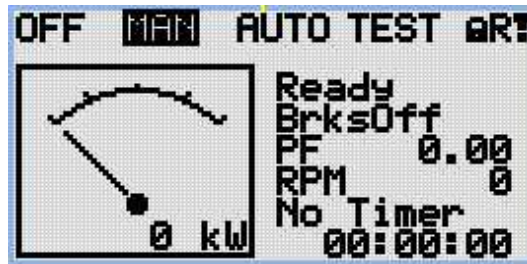


Note: Use Enter button to confirm the password or Page button to cancel entering the password.

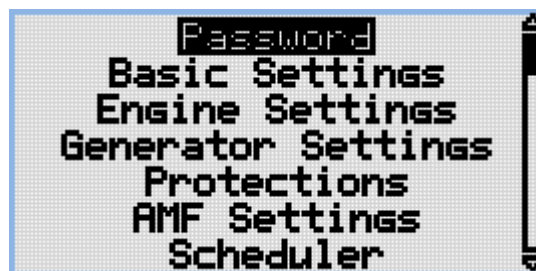


Note: In case that invalid password is entered, the controller shows Invalid password screen. Use Page button to go back to menu.

4.2 Change password



Note: From all measurement pages we can fluently go to the setpoint group page by pressing Page button.



Note: Use Up and Down button to select setpoint group Password.



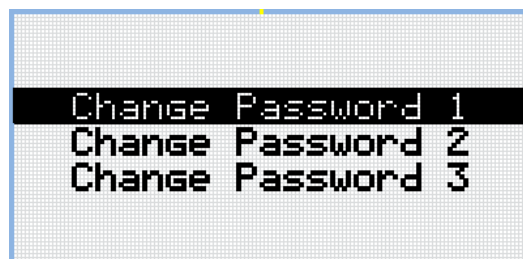
Note: Use Enter button to enter setpoint group Password.



Note: Use Up and Down button to select Change Password.



Note: Use Enter button to enter selected setpoint.



Note: Use Up and Down button to select required level of password.



Note: Use Enter button to enter selected setpoint.



Note: Use Up and Down button to set required value of password.



Note: Use Left and Right button to move between digits.



Note: After setting new password use Enter button to confirm adjusted password.



Note: Use Up and Down button to set required value of password again.



Note: Use Left and Right button to move between digits.

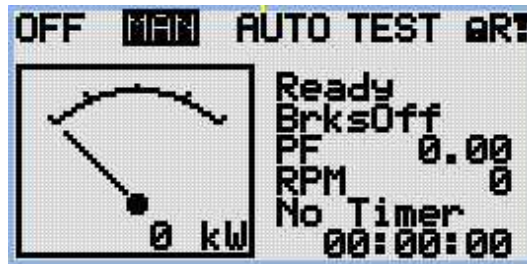


Note: After setting new password again use Enter button to confirm adjusted password or Page button to discard changes and to cancel changing password.

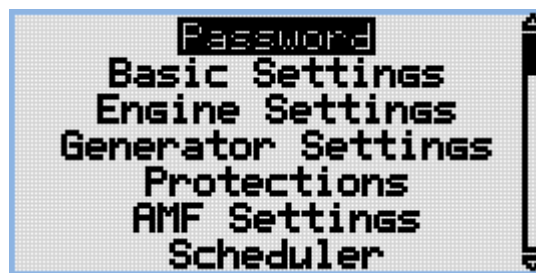


Note: Before changing the password controller has to be unlocked. In case that controller is locked, the controller shows Password required screen. In that case the password has to be entered before changing the password.

4.3 Log out from controller



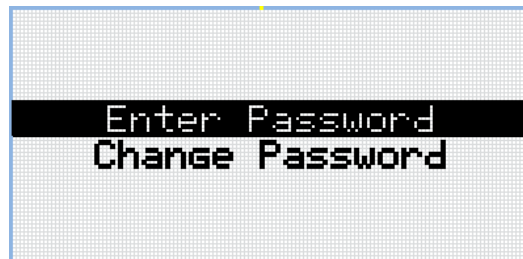
Note: From all measurement pages we can fluently go to the setpoint group page by pressing Page button.



Note: Use Up and Down button to select setpoint group Password.



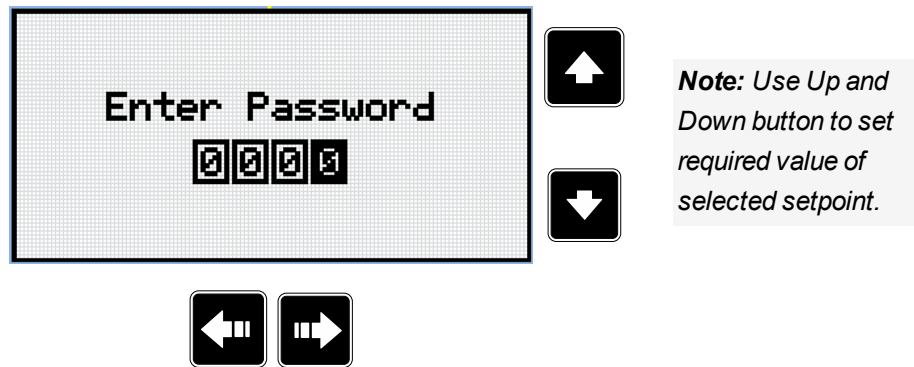
Note: Use Enter button to enter setpoint group Password.



Note: Use Up and Down button to select Enter Password.

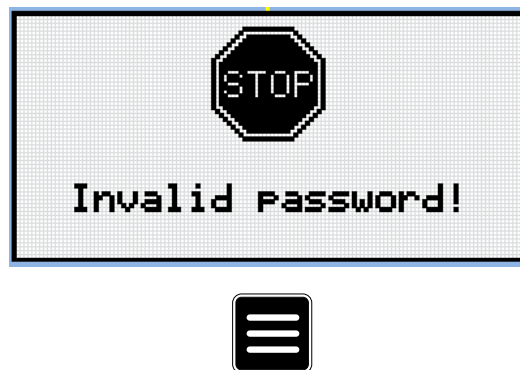


Note: Use Enter button to enter selected setpoint.



Note: Use Left and Right button to move between digits.

Note: Enter invalid password to log out from controller.



Note: In case that invalid password is entered, the controller shows Invalid password screen. Use Page button to go back to menu.

4.4 Lost Password

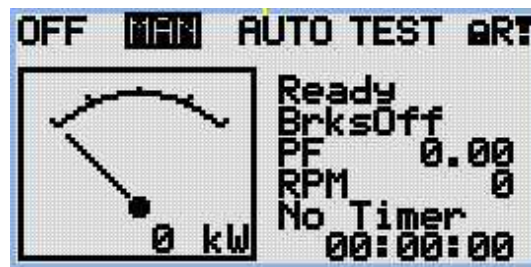
IMPORTANT: Display the information screen containing the serial number and password decode number as described in the chapter Information screen (page 24) and send them to your local distributor.

4.5 Password break protection

The controller password is protected against breaking by brute force. The protection works at every controller interface separately.

1. When an invalid password is entered 5 times after each other, independently on the time period elapsed between the attempts, the controller will be blocked after the 5th unsuccessful attempt for 1 minute at the particular interface.
2. While the controller is blocked it refuses any further attempts to enter password.
3. When unblocked again the controller accepts one attempt to enter password. If the password is incorrect again the controller will be blocked for 2 minutes.
4. Each further attempt to enter invalid password will double the blocking time, but maximum blocking time is 20 minutes.
5. When incorrect password is entered 100-times after each other the controller is blocked forever and the password reset procedure is required to unblock it.

5 Information screen



Note: On Main measurement screen press Enter and Page button together. Enter button has to be pressed first.



Note: Use Page button to move to the next page.



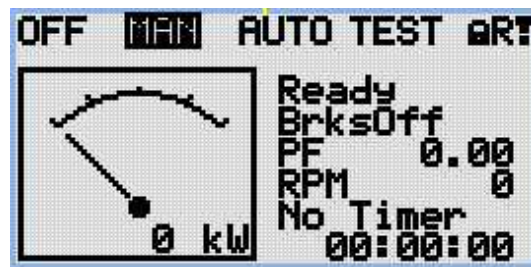
Note: Use Page button to move to the next page.

```
About Controller 2/2  
SW Version:    1.0.0.00  
HW Version:    1.0  
Serial:       12345678  
Pwd.Dec.:    1212345678
```

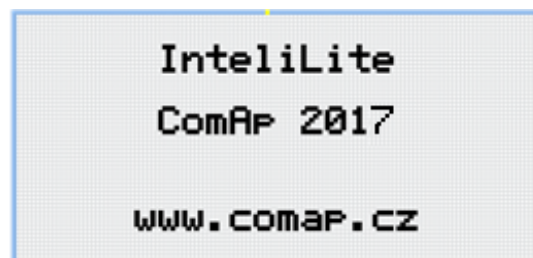


Note: Use Up button to move back to main measurement screen.

6 Language selection



Note: On Main measurement screen press Enter and Page button together. Enter button has to be pressed first.



Note: Use Page button to move to the next page.



Note: Use Page button to move to the next page.

```
About Controller 2/2
SW Version:    1.0.0.00
HW Version:    1.0
Serial:        12345678
Pwd.Dec.:     1212345678
```



Note: Use Page button to move to the next page.

```
LANGUAGES
Language1
Language2
Language3
```

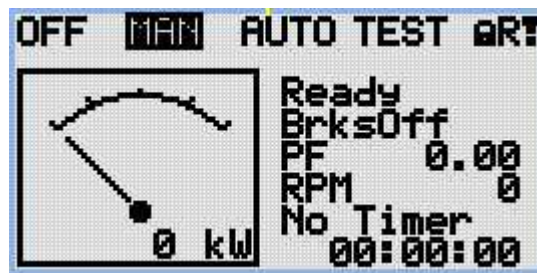


Note: Use Up and down button to select required language.



Note: Use Enter button to confirm selected language.

7 Display contrast adjustment



Note: On any measurement screen press Enter and Down button together for lower contrast.



Note: On any measurement screen press Enter and Up button together for higher contrast.



Note: After setting a contrast, no another action is needed.